

Institute of Actuaries of India

Subject SP7 – General Insurance Reserving and Capital Modeling

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Accounting concepts
- Going Concern : The enterprise will continue in operational existence for the foreseeable future. [0.5]
 - Accrual basis: Revenue and costs are recognised as they are earned or incurred, not as money is received or paid. [0.5]
 - Consistency: There is consistency of accounting treatment of like items within each accounting period and from one period to the next. [0.5]
 - Prudence and realisation: Revenue and profits are not anticipated (that is, must be realised), and provision is made for all known liabilities, whether the amount of these is known with certainty or is a best estimate in the light of the information available. [1]
 - Separate valuation of assets and liabilities: When determining the aggregate amount of any line item the enterprise must determine separately the amount of each individual asset or liability that makes up that item. [0.5]
- [3]**

ii)

- a. Accrual basis concept is non-complied with here. This is because the initial expenses are completely recognised in the first year itself as they are paid as there is no DAC concept. [2]
- b. 50% method for UPR would be a better approach in order to match the higher earning against higher initial expenses outgo in 1st year. This will remove the distortion which will arise in case the accounting for UPR is done on 1/365th method but at the same time DAC is not allowed. As otherwise 1/365th method of accounting is appropriate as it is consistent with Accrual basis concept when the premiums are recognised when they are earned. [2]
- c. The implications when 50% method is adopted for UPR
- UPR for policies which are more than 1 year old as at any valuation date will be zero.
 - Any claims arising in future when there will be no premium earning will distort the portfolio performance.
 - If the overall portfolio is profitable then the profits will be front loaded through this accounting which might go against the policyholder's interest.
 - Overall accounting results will be distorted, and judgement and further analysis has to be done to interpret the correct picture.
 - Such business may be prone to CAT perils as the insurance coverage period is long enough to be subjected to CAT events.

The implications when 1/365th method is adopted for UPR

- The accounting results will be distorted if DAC is not allowed
- This will be detrimental to the shareholders interest.
- Any claims arising early will significantly distort the financial results.
- If the overall portfolio is profitable, then the profits are back loaded and may not give the true picture. [0.5 mark for any point]

[3]

- d. Reinsurance arrangements can be made to avoid such distortions. [1 mark per point]

Long term quota share arrangements can be made through which the RI Ceding Commissions obtained can match the initial expenses paid for procurement.

This will avoid the distortion to a large extent because of no DAC being allowed in India.

- e. IFRS accounting has DAC concept inbuilt in it.

[3]

[1]

[11]

[14 Marks]

Solution 2:

- i) The key principles of investment strategy are

- Maximise returns subject to meeting all the contractual obligations [0.5]
- Assets should match liabilities by term, amount, nature and currency. [1]

If the classes of business are written with widely fluctuating claims, then there will be a need for liquidity.

[0.5]

[2]

- ii)

- a. Key issues are

- Prone to CAT events
- Going through a soft phase of insurance cycle which means profitability is under pressure
- Long term property insurance book and hence long-term exposure

In this scenario, a balance must be made between three aspects - liquidity, investment profitability and long-term nature.

Liquidity because loss making business and there is a high chance of a large CAT event loss.

[0.5]

Investment profitability because this will help make up the losses arising out of the insurance business.

[0.5]

Long term nature as the insurance portfolio is long term in nature.

[0.5]

Therefore, long term govt bonds which are liquid and marketable can be used for generating investment profits as well as meet the long-term liabilities.

[1]

In order to meet the immediate and short-term liquidity needs, adequate portion of investments can be made in Mutual funds and equity.

[0.5]

[3]

- b. Key issues are

- Medium sized company
- Having exposure to long tailed motor third party liability business
- Short tailed businesses are loss making

In this scenario, high proportion of liquidity must be maintained in the investment portfolio in order to meet the short-term liquidity needs as the short-tailed businesses are loss making.

[0.5]

This can be achieved by investing in liquid mutual funds.

[0.5]

In order to manage the long-term liabilities arising out of motor third party business long term government bonds are better assets which are liquid as well as marketable.

[1]

The long term govt bonds would also generate better investment returns.

[0.5]

Since the size is medium, not much free reserves would be available for investing in riskier equity investments.

[0.5]

[3]

- c. Key issues are
- Well capitalized which is not an issue actually
 - Recently started
 - Profitable business from group companies

Liquidity is not a concern here. Free reserves are expected to be high as mostly profitable business from group companies. High return investments are a better choice as risks can be taken with backing of free reserves. [1]

Long term investments generating more returns would be a better choice. [0.5]

This might be at the cost of having some duration mismatch but may not be a significant risk. This is because the company has no major liquidity needs. [1.5]

[3]

iii) ALM challenges can be minimized through the following reinsurance arrangements

- For products with high initial expenses any reinsurance arrangement which provides high ceding commissions matching the high level of initial expenses e.g. quota share etc. This will to a large extent reduce the liquidity risk arising out of the high initial payments.
- Proportional as well as non-proportion treaties which provide CAT cover with cash call arrangements with the reinsurers in case of large loss/CAT events. This will reduce the liquidity risk during the events of crisis.
- Reinsurance arrangements like time and distance deals (e.g. Loss Portfolio Transfers and Adverse Development Covers) which will take away the amount and timing mismatch risk from the reinsured books. This risk in relation to uncertainty in timing and amount of liabilities especially for long tailed business-like liability.
- Reinsurance Covers like stop loss provides certainty around the maximum outgo from a portfolio which then makes it much easier for managing risks within the ALM framework.

[1 mark for any point]

[3]

[14 Marks]

Solution 3:

i) The broad categories of Finite Risk Reinsurance are

- Pre-funded arrangements, whereby the insurer pays premiums into a fund held by the reinsurer (which earns interest), and claims are paid from that fund. [1]
- Post-funded arrangements, whereby the reinsurer pays the losses and the insurer pays back the losses over time. [1]

The specific types of finite risk reinsurance are [0.25 for any point subject to a maximum of 1]

- Time and distance policies
- Spread loss covers
- Financial quota share
- Structured finance
- Industry loss warranties

[3]

ii) The possible reinsurance arrangements under this scenario can be

- Time and Distance deals e.g. Adverse Development Cover or Loss Portfolio Transfer
- Mostly these are pre-funded instruments where a certain reinsurance premium is paid against the liabilities which will arise in future. Such liabilities constitute mainly outstanding claims, IBNR and IBNER claims. Since the motor third party businesses are long tailed in nature such

arrangements will reduce the level of uncertainty in terms of timing and amounts of future liabilities to a large extent. This will reduce the capital requirements. Also, the difference between the reinsurance premium paid and the liabilities taken off the book adds to the capital base of the reinsured. The difference arises because of the timing differences between the payment of reinsurance premium and liabilities which are paid off in future. The exact difference depends on the investment income opportunities available with the reinsurer and settlement practices followed by the reinsured. [2.5]

- Financial Quota share. Through this arrangement high ceding commissions can be obtained on which investment income can be generated. Due to reduction of volatility of claims the capital requirements also come down. Depending on the nature of the solvency capital structure in each jurisdiction, such Financial Quota Share arrangements can give further relief to the company in terms of reduction of solvency capital requirements. [2.5]

[5]

iii) Mack Model and Over-Dispersed Poisson Models

[1 mark for any point]

- For the time and distance deals like adverse development cover, it is important to fix the attachment points and upper limits based on the past claim's development pattern and the inherent risk involved in that. The attachment point is more driven by the reinsured requirement on how much to gain from the treaty in terms of solvency capital requirements whereas the upper limit is more driven by the uncertainty involved around the maximum claims to be ultimately paid to the policyholders. Normally the upper limits are fixed at 99.5% or 99% percentile depending on the VaR adopted by the reinsured. In order to derive these percentiles a full-blown distribution of outstanding and IBNR/IBNER claims is needed based on stochastic reserving techniques.

Since the reinsured company is large and old, it is expected that the paid claims development data is credible in addition to the incurred claims development data also. Since this is Motor Third Party liability business a substantial portion of claims development data will be in the form of OS claims.

When performing the stochastic reserving techniques on the paid claims development data, both Mach and ODP Bootstrap methods would work better as the possibility of both individual incremental claims and cumulative incremental claims getting negative is nil. Whereas when performing the stochastic techniques on the incurred claims development data, ODP Bootstrap method might break down as it cannot handle negative incremental claims both individual as well as cumulative. Therefore Mack is a better option as it can still handle individual negative incremental claims. [4]

[12 Marks]

Solution 4:

- i) Most latent claims have a "calendar year" development effect, that is, a similar claims development pattern for a calendar year rather than a development year. Thus a group of underwriting years can show development that results from the same underlying latent claims. [1]

Because of this, the earlier underwriting years cannot be easily used as a guide to how the later underwriting years might develop. Therefore chain ladder and other triangulation methods are not usually suitable for latent claims. [1]

[2]

- ii) Survival Ratio is the ratio of loss reserves to yearly paid loss average. [1]

It is often used to estimate the reserve adequacy for latent claims. Assuming average yearly paid losses remain constant, with no additional reserving, the survival ratio indicates how many years the reserves should last. [1]

[2]

iii)

- a. The yearly loss paid average is Rs. 1,00,00,000, where $1,00,00,000 = 800,00,000/8$

The survival ratio is given to be in the range of 14 to 16 in Dec 2010.

From the given data the total commutation offer as at Dec 2010 has reduced by 8,00,00,000.

Assumptions:

- Total ultimate liability remains the same as at Dec 2010 as well as at Dec 2018 and will remain the same in future also. [0.5]
- The yearly paid loss average will remain the same in future years as in the past. [0.5]

With the above assumptions,

The survival ratio as at Dec 2018 is expected to be in the range of 6 to 8 years (where $6 = 14-8$ and $8 = 16-8$). [0.5]

The implied survival ratio according to the revised commutation offer in Dec 2018 is $9,09,00,000/1,00,00,000 = 9.09$ [1]

Since 9.09 falls outside the range of 6 to 8, then the implied reserves strength in the commutation offer is substantially higher than that of the global benchmarks. The commutation offer therefore may result in more outgo for the reinsurer than the necessary amount. Therefore, recommend reject the offer and ask for further negotiations. [1.5]

[4]

- b. [0.5 for any point subject]

- Buy further traditional reinsurance like Stop Loss, etc.
- Keep the business in book and let it run off for a better result in the end. Globally the yearly average paid loss has been declining for such latent claims and hence it is better to keep the business and allow the ultimate liability to emerge in its book itself.
- Investment income may be a crucial factor in the case above.
- Buy Finite reinsurance like Capital Relief Quota Share, Time and Distance Deals, etc
- Cost of such reinsurance covers will be crucial.

[2]

[10 Marks]

Solution 5:

[0.5 mark for any point]

- i) New regulator may make changes to the regulatory environment as follows:
- a. They may make certain types of cover compulsory for policyholder to purchase or for insurers to provide e.g. motor third part liability, Household insurance in high risk flood areas
 - b. They may also change the amount of compulsory cover that has to be purchased, e.g. increase the compulsory coverage required for Motor third party property damage claims
 - c. They may restrict the type and amount of business that ABC may write.
 - d. They may change the capital regime in Siliconia e.g. introduce a new risk based capital calculation
 - e. ABS may be required to maintain a minimum level of solvency e.g. a minimum level of free assets

- f. They may introduce a new compensation scheme to protect policyholders should general insurance companies fail.
- g. They may change the way mechanism are funded e.g. through levies
- h. They may change regulatory reporting requirements in Siliconia e.g.:
- i. Increased the level of disclosure required
- j. Require all the general insurers to obtain an actuarial opinion to support their claims reserves.
- k. Imposing limits on the premium that can be charged e.g. max or min premium
- l. Restricting the information that may be used in underwriting or premium rating e.g. prohibition the use of gender
- m. Require ABC to file or publish its premium rates before they can be used
- n. Restricting the countries where it can write business
- o. ABC may be required to deposit assets to back its claims reserves
- p. They may prescribe the basis to be used to calculate premium , assets values, and liabilities to demonstrate solvency.
- q. They may place restriction on the type or amount of certain assets allowed to demonstrate solvency
- r. They may restrict type of reinsurance that ABC may use.
- s. They may introduce new regulations wrt treating customer fairly e.g. by introducing a cooling off period.
- t. They may change the rules around how insurance products may be sold and by whom [5]

ii) Advantage

[0.5 marks for any point]

- a. Simplicity – Regulator won't need to review and approve every company's internal capita model
- b. Cost saving – resulting in cost saving to both regulator and insurer, who don't need to build such models,
- c. The regulator would need to employ fewer staff and require less expertise
- d. As motor and household insurance are compulsory, there should not be any issues regarding policyholders who are unable to obtain cover e.g. high flood risk areas
- e. Confidence in the insurance industry may grow as the regulator would effectively become the insurer of last resort
- f. With both products becoming compulsory, the total claims risk and hence capital would be fixed, so it may be intuitive for capital to be manged centrally
- g. The capital requirement of the fund would effectively be diversified across the whole market, meaning
- h. Less capital is needed in aggregate, which should result in lower premiums for policyholders
- i. Lower capital requirement can create competitive advantage
- j. Fund should have greater investment freedom and benefit from more economies of scale
- k. Insurers could hold gross reserves and reinsurance can be negotiated at fund level
- l. Regulator won't need to worry about errors in capital model
- m. Contribution to fund could be combined with other levies
- n. As regulator is controlling fund , it is determining how much capital is being held by insurance market in aggregate [5]

Disadvantage

[0.5 marks for any point]

- a. Insurer will pay less focus on the risk they were writing as capital would only be a function of the number of policies written,
- b. No incentive for insurer to manage risk such as aggregation or correlations, riskier policies become more attractive
- c. Insurers may be more likely to fail resulting in more calls on the central fund
- d. Capital regime would be out of line with international standards
- e. If the levy were to vary according to the premium, then insurers may deliberately under price and regulator would need to monitor premium rates
- f. The regulator will need to determine the size of the fund and the size of the levy
- g. It may not have necessary skills, expertise and data to do this
- h. The regulator could request additional data from insurers to calculate the capital required but this would increase the cost of regulation, may offset the cost saving made elsewhere
- i. The regulator would need to review the best estimate reserves held by insurers which could create time and cost implications
- j. Insurers are less likely to invest cost and time in risk management procedures which might increase -
 - 1. Overall level of risk in the market
 - 2. Overall industry capital requirement i.e. size of fund
 - 3. The size of the levies to each insurer
 - 4. Policyholders premium
- k. Likely to be knock on adverse effects to reinsurance market
- l. if insurers don't need to purchase reinsurance
- m. Reinsurance is purchased centrally by the fund thus reducing demand
- n. Transition from present regime to new fund is complicated, time consuming and expensive
- o. Not all policyholders want to buy the product e.g. some may self insure and others may not afford the premium
- p. As the fund covers the entire market risk, if it were to fail, the insurance market would collapse, whereas the failure of an individual insurer would not necessarily bring down the entire market.

[5]

[10 Marks]

Solution 6:

i)

[0.5 mark for any point]

- a. We might expect a reasonable proportion of total claims cost to arise from large individual claims greater than 1500000
- b. Large claims likely to arise from bodily injury rather than due to property damage
- c. The likelihood of large claims is usually higher for commercial due to higher mileage driven, , although it will depend upon the experience of the drivers
- d. The likelihood of large claims is usually higher for young drivers who may generally have non comprehensive rather than comprehensive insurance
- e. The likelihood of large claims is increasing due to court awards and general litigiousness.
- f. Catastrophes may arise from a motorway pile up or weather events such as floods
- g. The potential for accumulation of risk is greater for commercial motor, although these are likely to have less impact on overall claims costs than large individual claims.

[3]

- ii) Claims data adjustment [0.5 mark for any point]
- If left unadjusted in the data, large claims might distort the projection of outstanding claims reserves.
 - This is the case if large individual claims have different claims development pattern to non large claims and the mix of non large and large claims varies from year to year
 - Leaving large claims in the aggregate data could then result in unstable chain ladder development factors and average development factors for each development year might be distorted by unusually high or low large loss experience in recent years.
 - Even when averages are not distorted, applying an average chain ladder development factor might be inappropriate for those years of account with unusually high or low large loss experience.
 - Catastrophes can cause a similar problem to individual large claims, although the various individual claims arising from a catastrophe may develop at a similar speed to non Catastrophes claims, they may bias the average date of occurrence.
 - Claims resulting from a storm and flood catastrophes tend to be reported very quickly and therefore distort the reporting patterns, whereas the claims from subsidence catastrophes tend to be reported slowly
 - Splitting of such claims leads to greater accuracy in modelling
 - The inflationary effect on a large claim is likely to be different to that on smaller claims
 - If left unadjusted in aggregate data, large claims would unduly dominate the experience of the risk group and might lead to inequitable pricing, which in turn might lead to anti selection.
 - This is particularly relevant for rating cells/risk groups with lower volumes of data, eg 80 year old drivers for private motor insurance
 - This could create non-competitive premiums. [4]

- iii) Ways of defining and treating large claims [2 marks for any valid approach and description]
There are different definitions of a large individual claims

Different approaches include:

- Do not extract large claims from the data
 - Simple and quick
 - Fairly stable if large claims experience has been fairly stable from year to year
 - Ensure reasonable allowance for unreported large claims
 - May result in over/under estimation of IBNR if large loss experience is not stable
 - Does not recognise trends in large claims experience
- Extract the whole of each large claim and associated history if its incurred claim amount exceeds a certain threshold
 - Non large claims triangles is not distorted by past history of large claims
 - Will need to restate the history of non large triangles each year as non large claims become large
 - Difficult to reconcile with last years data
 - Difficult to allow for claims currently classified as non large to become large
- "once large always large" i.e even if Incurred claim amount for a loss falls back below threshold, still treat as "large".

- Reduces the need to amend history of non large triangles each year
 - Recognises the potential for large claims to become non large and therefore avoids over estimation of reserves for large loss
 - May distort any large claim average cost analysis
4. Only extract claim from the point that it becomes large, ie history of the claim before it was large remains in the aggregate data
- No need to amend history of non large triangles each year
 - May be sharp reduction in claims in non large triangles from one development year to next
 - Development factor that rely too heavily on such an instance would result in an optimistic non large claims estimate
5. Apply indexing to large claims definition
- Ensures that large claim definition maintains real value over time
 - If there was no indexation, there would be very few claims extracted from early years of account compared to later years and this would reduce the reliability of any development analysis
 - Indexation introduces complexity
 - Inflation is hard to measure
6. Only extract part of each large claims that is in excess of threshold.
- The non-large aggregate claims history does not change over time
 - If the threshold is in line with excess point for excess of loss reinsurance then reinsurance IBNR can be identified more easily
 - Might be harder for systems to extract the excess over the threshold.

[12]

[19 Marks]

Solution 7:

- i) Latent claims - Claims arising from a risk not specifically allowed for by the underwriter when selling or pricing the original policy. [1]
The term is sometimes loosely used to describe any claim where there is a long delay between the original exposure and the manifestation of the claim. [1]
[2]
- ii) Examples include: [0.5 mark for any example]
1. Industrial deafness,
 2. Asbestos related diseases-including asbestosis, mesothelioma, pleural plaques
 3. pollution and chemical contamination
 4. occupational hazards - Vibration white finger, noise induced deafness, repetitive strain injury
 5. Sexual abuse claims
 6. Tobacco
 7. Product claims and Pharmacological treatment

Insurance product affected include:

[0.5 mark for any product]

1. Employer's liability and workers compensation

2. Product liability (affected by tobacco and pharmacological claims)
3. Environmental cover(affected by tobacco and pharmacological claims)
4. Public liability (affected by pollution and abuse claims)
5. Comprehensive general liability

[4]

iii) Allowing for latent claims when modelling capital requirements

Reasons

[2]

The main reason for considering latent claims when modelling capital requirements is the uncertainty in timing and emergence of these claims as well as their uncertain size and frequency. This means that the extra capital needed could be considerably high, both from a regulatory and an internal view.

The most likely scenario (driven by solvency II guidelines), required capital should be calculated to keep the probability of insolvency to below 0.5% over a one year time horizon.

These parameters can be built into model when allowing for latent claims.

Consider the views of rating agencies, who would expect capital to be available for these potential events.

Methods

[3]

Deterministic or stochastic models can be used for latent claims.

Deterministic methods may be too simplistic to model such uncertain and complex events, although they could at least be supplemented with some stress and scenario testing.

Stochastic models may have limited use as the uncertainty of latent claims would make such models difficult to parameterise.

These could therefore be substantial model or parameter error.

For example, bootstrapping methods require past data which may not be available in sufficient quantities.

Stochastic methods are often criticised as under-estimating the true volatility of events.

Stochastic methods are more feasible for a large company. The methodology and results should be communicated carefully.

Distribution used in the reserving model may be a good starting point within the capital model, but the extremes of the distribution need an extra level of scrutiny. Scaling factors could be used but should not be excessive.

Another methods could be frequency/severity methods, a statistical method or a combination of two.

Allow for reinsurance. It may be that reinsurance arrangements would cover many latent claims, although possibility of reinsurer default and disputes over policy wording should also be factored in.

Allow for reinsurance costs escalating.

Risks

[2]

When considering reserve risk, there are many components of the outstanding claim reserves which may be involved. i.e. reported claims, IBNR, IBNER, reopened claims and claim handling expenses, with allowance for reinsurance recoveries.

For latent claims, the risk of under-reserving could be substantial.

Risk allowances within the reserving risk module should therefore be augmented to allow for this.

The risk of under-pricing is also an issue, as any loading to premium rates to allow for potential latent claims can only be small given market considerations.

However, it is perhaps less important than reserving risk as there are so many other factors influencing prices.

Market risk can arise from worsening inflation affecting claims or even exchange rates, depending on the location of claims.

Operational risk should be minimal in respect of latent claims.

Diversification/Correlation

[1]

There will probably be correlation between latent claims and other aspects of the capital model. For example, presence of latent claims in a particular year would indicate exposure to latent claims in adjacent years.

Other examples of where correlation could occur are as a result of legislative changes (e.g. relating to compensation funds) and policy wording application.

Practical considerations

[2]

Data will be minimal when considering any aspect of modelling latent claims, including the building, calibration, and validation of the model.

Calibration will require the use of judgement and benchmarks.

Model validation should include sensitivity testing and stress testing , to assess the suitability of assumptions such as inflation and propensity to claim.

However, this is again subjective , since many things can change such as legislation, court rulings, development in disease manifestation and diagnosis.

Validation of model (both methodology and assumptions) will be difficult given uncertain nature of latent claims. Most of assumptions will be highly subjective and based on minimal data. [10]

[16 Marks]
